THERMODYNAMICS

Course Code : 23A03301 | Handwritten Study Notes

UNIT – IV : Pure Substances & Steam Properties

★ Note: Syllabus: Pure substances, P-V-T surface, T-s & h-s diagrams, Mollier charts, phase transformations, triple point & critical state, dryness fraction, Clausius-Clapeyron equation, property tables, thermodynamic processes of steam, steam calorimetry

1. Pure Substance

A pure substance has a homogeneous and invariable chemical composition throughout its mass, even if the phase changes.

- Examples: water (ice, water, steam), nitrogen, carbon dioxide, refrigerant R-134a, air (gas phase only, as a mixture of fixed composition).
- A mixture of water and steam is pure; a mixture of liquid air and gaseous air is NOT pure (composition differs in phases).
- Simple compressible substance: only P-V work; two independent properties fix the state.
- Gibbs phase rule: $F = C - P + 2$ (F = degrees of freedom, C = components, P = phases). Single-phase pure substance: $F = 2$; two-phase: $F = 1$; triple point: $F = 0$.

2. Phase Transformations of Water at Constant Pressure

Heat 1 kg of ice/water in a piston-cylinder at constant pressure and watch T and v:

→ Sub-cooled (compressed) liquid – $T < T_{sat}$; heating raises T. Sensible heat.

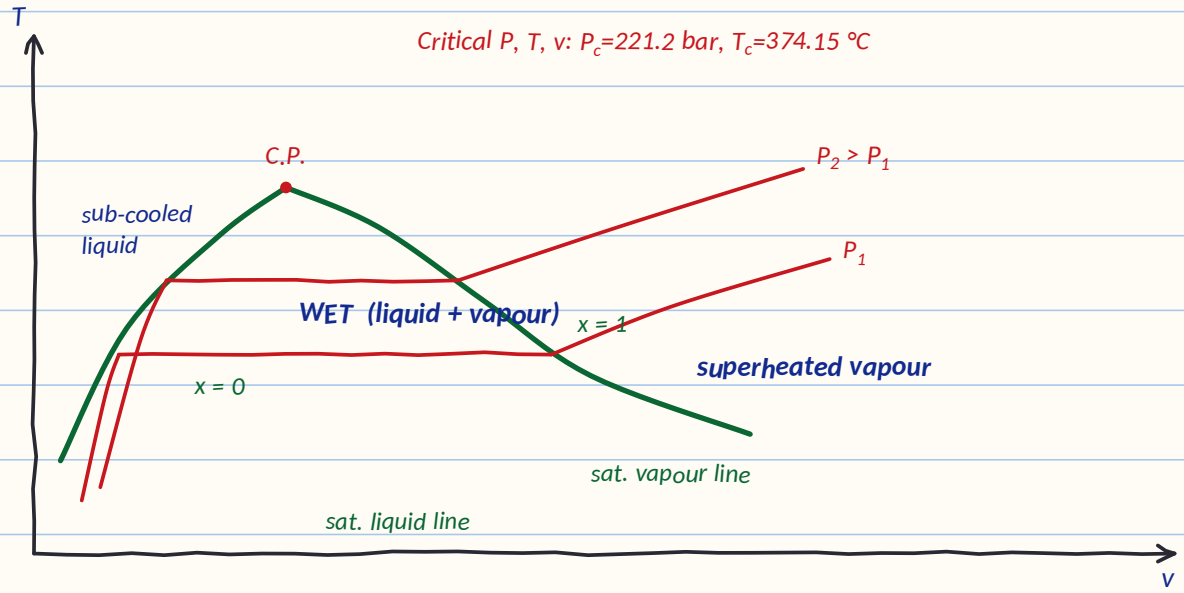
→ Saturated liquid – liquid at T_{sat} (about to evaporate). Properties with

subscript f.

→ Wet (saturated mixture) – liquid + vapour coexist at constant T_{sat}; heat added is latent heat; dryness fraction x rises 0 → 1.

→ Dry saturated vapour – x = 1; last drop evaporated. Subscript g.

→ Superheated vapour – $T > T_{sat}$; heating raises T again.

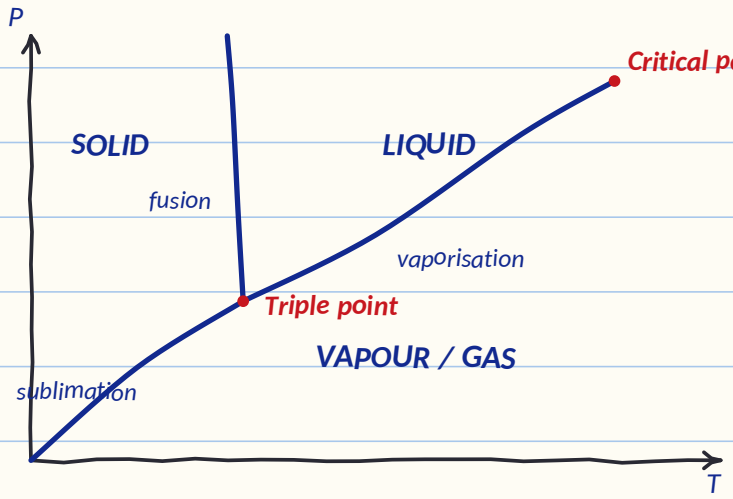


- As pressure increases, T_{sat} rises and the horizontal evaporation line (h_{fg}) becomes shorter; at the critical point it shrinks to a point.
- Degree of superheat = $T_{sup} - T_{sat}$. Degree of sub-cooling = $T_{sat} - T$.

3. Triple Point and Critical Point

- Triple point: the unique state at which solid, liquid and vapour phases coexist in equilibrium. Water: 0.01 °C, 0.00611 bar (0.6113 kPa).
- Critical point: the state at which the liquid and vapour phases become identical – no distinct evaporation, $h_{fg} = 0$. Above it, a fluid is "supercritical".

Water critical constants: P_c = 221.2 bar, T_c = 374.15 °C, v_c = 0.00317 m^3/kg



Triple point of water:
0.01 °C , 0.00611 bar

Critical point of water:
374.15 °C , 221.2 bar

Triple point is a fixed point
($F = C - P + 2 = 0$)

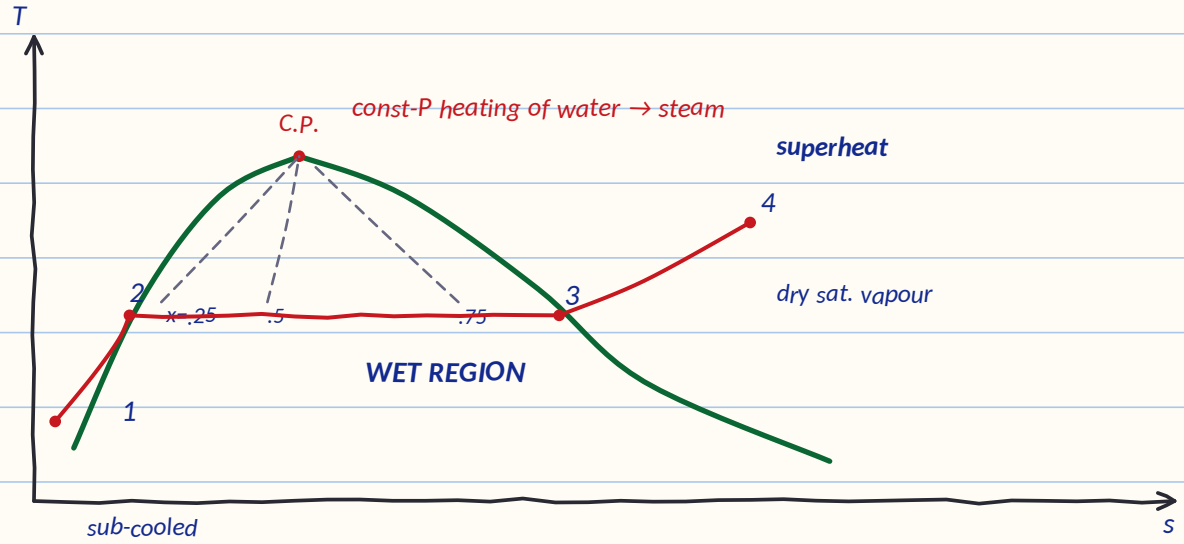
- P-T diagram: sublimation curve (solid ⇌ vapour), fusion curve (solid ⇌ liquid), vaporisation curve (liquid ⇌ vapour, ends at the critical point).
- For water the fusion curve slopes backward (ice expands on freezing, melting point falls as P rises); for most substances it slopes forward.

4. P-v-T Surface

The equilibrium states of a pure substance form a surface in three dimensions (P, v, T).

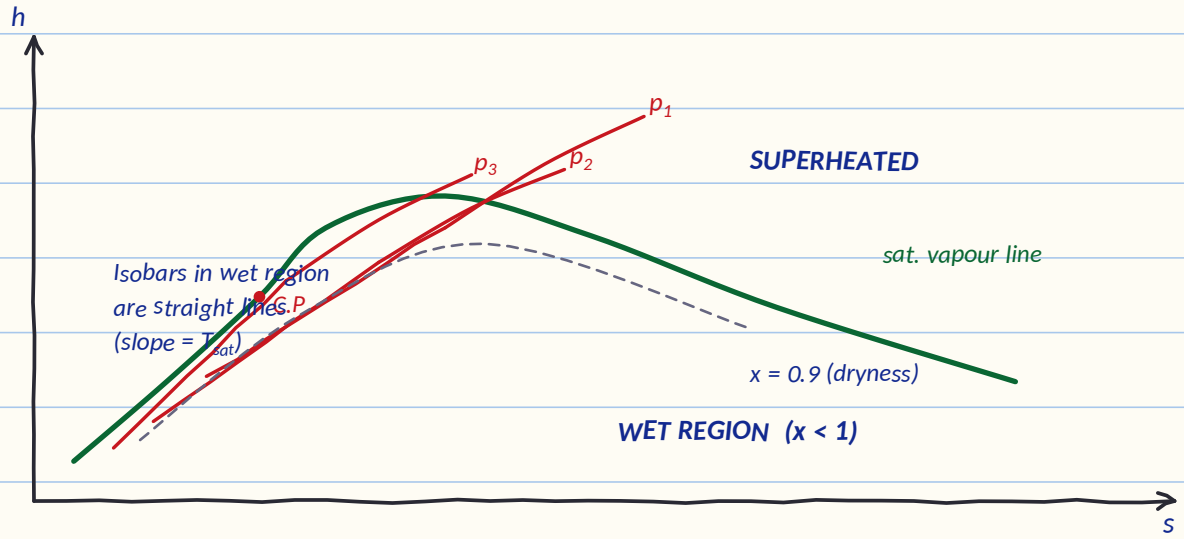
- Single-phase regions (solid, liquid, vapour) are curved surfaces; two-phase regions (solid-vapour, solid-liquid, liquid-vapour) are ruled surfaces parallel to the v-axis because P and T are fixed during phase change.
- The triple line is a line of constant P and T where three phases coexist; its projection on the P-T plane is the triple point.
- Projections: P-T diagram (phase boundaries), P-v diagram, T-v diagram – the dome shape of the liquid-vapour region is seen in the last two.
- Substances that contract on freezing (most) vs expand on freezing (water, antimony, bismuth) have slightly different surfaces.

5. T-s Diagram of Water



- Left of the critical point the saturated liquid line ($x = 0$); right the saturated vapour line ($x = 1$); inside the dome = wet region.
- Constant pressure heating 1→2→3→4: sub-cooled → saturated liquid → dry steam → superheated steam.
- Constant dryness lines ($x = 0.25, 0.5, 0.75$) converge at the critical point.
- Area under a reversible curve = heat transferred ($\int T\,ds$). Reversible adiabatic = vertical line.
- Latent heat of vaporisation $h_{fg} = T_{sat}\, s_{fg}$ (area under the horizontal line).

6. Mollier (h-s) Diagram



- Mollier chart plots enthalpy (ordinate) against entropy (abscissa). Isentropic process = vertical line; throttling = horizontal line (h const).
- Constant-pressure lines: straight lines with slope = T_{sat} in the wet region (they diverge upward), curves in the superheated region.
- Constant-dryness lines, constant-temperature lines (superheat region) and the saturation line are given.
- Advantage: enthalpy drop in turbines and nozzles read directly as a vertical distance. Practical working region is mostly wet and superheated steam.

7. Dryness Fraction and Properties of Steam

★ Note: Dryness fraction x = mass of dry vapour / total mass of mixture = $m_g/(m_f + m_g)$. Wetness fraction = 1 − x. Liquid: x = 0; dry saturated: x = 1.

$$v = v_f + x\, v_{fg} \approx x\, v_g$$

$$h = h_f + x\, h_{fg} \qquad u = u_f + x\, u_{fg}$$

$$s = s_f + x\, s_{fg} \qquad h_{fg} = h_g - h_f$$

- Superheated steam: $h = h_g + c_{ps}(T_{sup} - T_{sat})$, $c_{ps} \approx 2.1$ kJ/kg·K (approx.).
- Sub-cooled liquid: $h \approx h_f$ at the given T, $v \approx v_f$ (almost incompressible).
- Internal energy: $u = h - P\,v$. Enthalpy of vaporisation decreases with pressure and is 0 at the critical point.

Property table (short extract – saturated water)

P (bar)	T_s °C	h_f	h_{fg}	h_g	v_g m³/kg	s_g
1	99.6	417.5	2257.9	2675.4	1.694	7.359
10	179.9	762.6	2013.6	2776.2	0.1944	6.586

8. Property Tables

- Saturation table (temperature based): gives P_{sat}, v_f, v_g, u, h, s at a given T. Saturation table (pressure based): gives T_{sat} etc. at a given P.
- Superheated table: at a pressure, lists v, u, h, s for various temperatures above T_{sat}.
- Compressed-liquid table: used at high pressure, otherwise approximate with saturated liquid at the same T.
- Decide the region first: compare given v or h with v_f, v_g (or h_f, h_g). $v_f < v < v_g \Rightarrow$ wet, so use x.
- Use linear interpolation between table rows when the exact value is not tabulated.

9. Clausius–Clapeyron Equation

Gives the slope of the saturation (vaporisation) curve on a P-T diagram. Starting from the Maxwell relation $(\partial P/\partial T)_v = (\partial s/\partial v)_T$, in the two-phase region P depends on T only:

$$\frac{dP}{dT} = \frac{s_{fg}}{v_{fg}} = \frac{h_{fg}}{T\,v_{fg}} \qquad \text{(Clapeyron)}$$

- For liquid-vapour: v_f negligible, vapour ideal $\Rightarrow v_{fg} \approx RT/P$:

$$\ln\left(\frac{P_2}{P_1}\right) = \frac{h_{fg}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \qquad \text{(Clausius–Clapeyron)}$$

- Applies to fusion ($dP/dT = h_{if}/(T\,v_{if})$, negative for water) and sublimation too.
- Example: at 100 °C, $h_{fg} = 2257$ kJ/kg, $v_{fg} = 1.672$ m³/kg $\Rightarrow dP/dT = 2257\times10^3/(373.15 \times 1.672) = 3.62$ kPa/K.

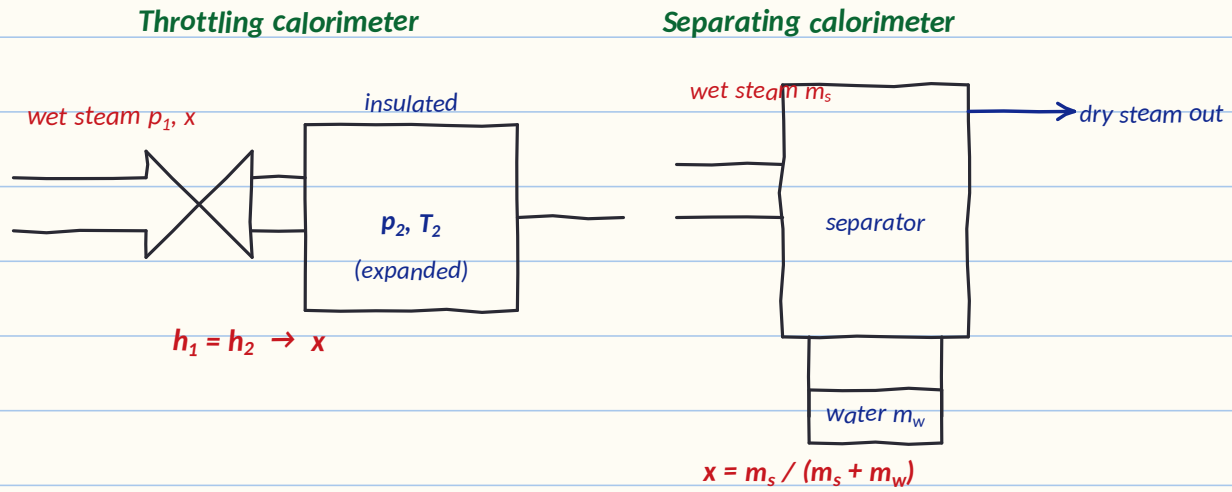
10. Thermodynamic Processes of Steam

Process	Work W	Heat Q	Remarks
Constant volume	0	u2 − u1	v1 = v2, find x2 or T2
Constant pressure	P(v2−v1)	h2 − h1	boiler, condenser
Isothermal	Q − Δu	T(s2 − s1)	wet: also const P
Isentropic (rev. adiabatic)	u1 − u2	0	turbine, nozzle; s1 = s2
Throttling	0	0	h1 = h2, irreversible
Polytropic P v^n = C	(P1v1−P2v2)/(n−1)	Δu + W	n=1.3 superheat, 1.135 wet

- Isentropic expansion in turbine: from state 1 draw a vertical line on h-s chart to P_2; ideal work $w = h_1 - h_2$. Actual work smaller; isentropic efficiency $\eta = (h_1 - h_{2a})/(h_1 - h_{2s})$.
- Throttling: steam flows through a restriction, P drops, h constant, entropy increases, T may fall or the steam becomes superheated.

11. Steam Calorimetry

Measures the dryness fraction x of wet steam in a pipe / boiler.



Throttling calorimeter

- Wet steam of pressure P_1 is throttled (h const) to a lower pressure P_2 so that it becomes slightly superheated; T_2, P_2 are measured.

$$h_1 = h_2 \Rightarrow h_{f1} + x\, h_{fg1} = h_{g2} + c_{ps}(T_2 - T_{sat2})$$

$$x = [h_{g2} + c_{ps}(T_2 - T_{sat2}) - h_{f1}] / h_{fg1}$$

- Limitation: steam must be fairly dry (x > about 0.95) so that it is superheated after throttling; otherwise use a separating calorimeter first.

Separating calorimeter

- Steam sample is passed through a separator; the moisture is collected and weighed; remaining dry steam is condensed and weighed.

$$x = m_s/(m_s + m_w) \quad (m_s\text{: dry steam, } m_w\text{: separated water})$$

- Limitation: separates only part of the moisture – inaccurate for very dry steam.

Combined (separating + throttling) calorimeter

- Separating calorimeter first removes most moisture (x_1); the throttling part finds the dryness x_2 of what remains. Overall dryness $x = x_1\, x_2$.

12. Solved Numerical Problems

Problem 1 : properties of wet steam

Steam at 10 bar, $x = 0.9$. Find v, h, s, u.

$$v = 0.9 \times 0.1944 \approx 0.175\ m^3/kg\ ;\ h = 762.6 + 0.9 \times 2013.6 = 2574.8\ kJ/kg$$

$$s = 2.138 + 0.9 \times 4.448 = 6.141\ kJ/kg\cdot K$$

$$u = h - Pv = 2574.8 - 1000 \times 0.175 = 2400\ kJ/kg$$

Problem 2 : dryness fraction from volume

A rigid vessel of 0.3 m^3 holds 2 kg wet steam at 10 bar. Find x.

$$v = 0.3/2 = 0.15\ m^3/kg\ ;\ x = (v - v_f)/(v_g - v_f) = (0.15 - 0.001127)/(0.1944 - 0.001127) = 0.77$$

Problem 3 : heat to make steam

Heat required to convert 1 kg water at 30 °C to dry saturated steam at 1 bar ($h_{f30} = 125.7$ kJ/kg).

$$q = h_g - h_{f30} = 2675.4 - 125.7 = 2549.7\ kJ/kg$$

Problem 4 : superheated steam enthalpy

10 bar, 250 °C ($T_{sat} = 179.9$ °C): $h \approx 2776.2 + 2.1 \times (250 - 179.9) = 2923$ kJ/kg (approx.)

Problem 5 : isentropic turbine

Dry saturated steam at 10 bar expands isentropically to 1 bar. Find work.

$$s_2 = s_1 = 6.586 \Rightarrow x_2 = (6.586 - 1.3026)/6.0568 = 0.872$$

$$h_2 = 417.5 + 0.872 \times 2257.9 = 2387\ kJ/kg$$

$$w = h_1 - h_2 = 2776.2 - 2387 = 389\ kJ/kg$$

Problem 6 : throttling calorimeter

Steam at 10 bar is throttled to 1 bar, 110 °C ($h_2 \approx 2696$ kJ/kg). Find x.

$$x = (2696 - 762.6)/2013.6 = 0.96$$

Problem 7 : separating calorimeter

2 kg sample, 0.3 kg water separated ⇒ dry steam 1.7 kg ⇒ $x = 1.7/2 = 0.85$

13. Heat Added at Each Stage of Steam Formation

Stage	Heat per kg	Remarks
Water 0 °C → T_s	$h_f = c_{pw}T_s \approx 4.187\,T_s$	sensible heat
Evaporation at T_s	$x\,h_{fg}$	latent heat (wet steam)
Complete evaporation	h_{fg}	dry saturated steam, $h_g = h_f + h_{fg}$
Superheating	$c_{ps}(T_{sup} - T_s)$	$h_{sup} = h_g + c_{ps}\Delta T$

- Latent heat is spent in two ways: external work of evaporation $P(v_g - v_f)$ to push back the atmosphere, and internal latent heat = $h_{fg} - P(v_g - v_f)$ to separate the molecules.

  At 1 bar: $P\,v_{fg} = 100 \times (1.694 - 0.001) = 169.3$ kJ/kg

  Internal latent heat = 2257.9 − 169.3 = 2088.6 kJ/kg

- Total heat of steam from water at 0 °C: wet $h_f + x\,h_{fg}$; dry h_g; superheated $h_g + c_{ps}(T_{sup} - T_s)$.
- Specific volume: $v \approx x\,v_g$ (v_f negligible except near the critical point); superheated steam behaves approximately like a gas: $P\,v = 0.4615\,T$.

14. Real Gas Behaviour (Link with Pure Substances)

- At low P and high T a gas follows the ideal gas law $Pv = RT$. Near the dome it departs strongly.
- Compressibility factor $Z = P\,v/(R\,T)$. $Z = 1$ for ideal gas; $Z < 1$ where attraction dominates; $Z > 1$ at very high pressure.
- van der Waals equation: $(P + a/v^2)(v - b) = R\,T$; a corrects for intermolecular attraction, b for finite molecular volume.

$$a = 27\,R^2\,T_c^{\,2}/(64\,P_c) \qquad b = R\,T_c/(8\,P_c) \qquad Z_c = 3/8 = 0.375$$

- *Law of corresponding states*: all gases at the same reduced $P_r = P/P_c$ and $T_r = T/T_c$ have nearly the same $Z \Rightarrow$ generalized compressibility chart.
- Other equations: Redlich–Kwong, Beattie–Bridgeman, virial expansion $Z = 1 + B/v + C/v^2 + \ldots$

15. More Solved Problems

Problem 8 : constant pressure heating of wet steam

1 kg steam at 5 bar, $x = 0.8$ is heated at constant pressure until dry saturated ($h_{fg} = 2107.4$, $v_g = 0.3749\ m^3/kg$).

$$Q = (1 - 0.8) \times 2107.4 = 421.5\ kJ$$

$$v_1 \approx 0.8 \times 0.3749 = 0.300\ ; \quad W = P(v_g - v_1) = 500 \times 0.075 = 37.5\ kJ$$

$$\Delta u = Q - W = 421.5 - 37.5 = 384\ kJ$$

Problem 9 : external work of evaporation

At 10 bar: $P(v_g - v_f) = 1000 \times (0.1944 - 0.001127) = 193.3\ kJ/kg$

Internal latent heat $= 2013.6 - 193.3 = 1820.3\ kJ/kg$

Problem 10 : throttling process

Steam throttled from 10 bar to 1 bar, h constant. If $x_1 = 0.96$, $h_1 = 762.6 + 0.96 \times 2013.6 = 2695.9\ kJ/kg$.

h_g at 1 bar $= 2675.4 < 2695.9 \Rightarrow$ steam leaves superheated ; $T_2 \approx 100 + (2695.9 - 2675.4)/2.0 \approx 110\ °C$

Problem 11 : turbine isentropic efficiency

Ideal $\Delta h = 389\ kJ/kg$ (Problem 5); actual $\Delta h = 330\ kJ/kg$

$$\eta_{isen} = 330/389 = 0.85 = 85\ \%$$

16. Glossary & Short Questions

→ *Saturation temperature* – temperature at which a liquid boils at a given pressure; rises with P.

→ *Sensible heat* – heat that changes temperature; *latent heat* – heat that changes phase at constant T and P.

→ *Why is the critical point special?* Liquid and vapour properties coincide; no latent heat; beyond it the fluid is supercritical.

→ *Why do isobars in the wet region on T-s become horizontal?* P and T are both constant during phase change.

→ *Why does steam leave a throttle valve at lower temperature?* Enthalpy constant, pressure falls – saturation temperature drops, steam may become superheated.

→ *Why use the combined calorimeter?* Throttling alone fails for very wet steam; separating alone misses fine moisture.

17. Quick Revision & Important Questions

- Pure substance: fixed composition. Triple point F = 0. Critical point: h_{fg} = 0.
- $h = h_f + x\, h_{fg}$; $v = x\, v_g$; $s = s_f + x\, s_{fg}$. Throttling: h const. Isentropic: s const.
- $dP/dT = h_{fg}/(T\, v_{fg})$. Isobars in wet region are straight on T-s and h-s.

→ Draw the P-T and T-v diagrams of water and mark triple and critical points.

→ Explain the formation of steam at constant pressure with a T-s diagram.

→ Derive the Clausius–Clapeyron equation.

→ Describe the throttling and separating calorimeters. Derive the formula for x.

→ Explain the Mollier chart and its advantages.

~ End of Unit IV ~